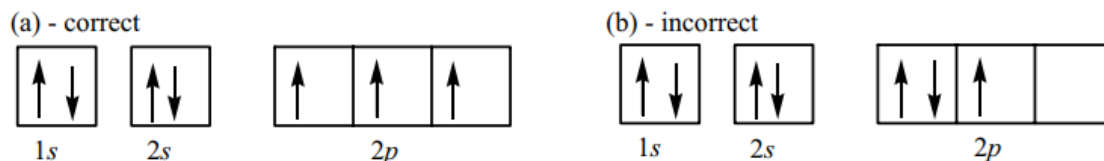


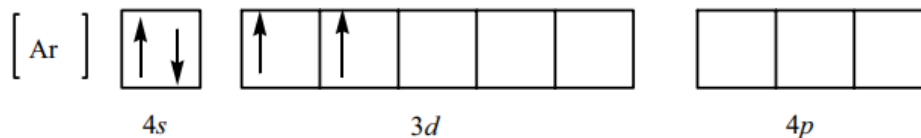
- 8.6 The quantum number m_s relates to just the electron; all the others describe the orbital.
- 8.9 Shielding occurs when inner electrons protect or shield outer electrons from the full nuclear attractive force. The effective nuclear charge is the nuclear charge an electron actually experiences. As the number of inner electrons increases, the effective nuclear charge decreases.
- 8.11 a) The $l = 1$ quantum number can only refer to a p orbital. These quantum numbers designate the $2p$ orbitals, which hold a maximum of **6** electrons.
 b) There are five $3d$ orbitals, therefore a maximum of **10** electrons can have the $3d$ designation.
 c) There is one $4s$ orbital which holds a maximum of **2** electrons.
- 8.13 a) **6** electrons can be found in the three $4p$ orbitals, 2 in each orbital.
 b) **2** electrons can have the 3 quantum numbers given for one of the $3p$ orbitals with the difference between the two being in the m_s quantum number.
 c) **14** electrons can be found in the $5f$ orbitals ($l = 3$ designates f orbitals).
- 8.16 Hund's rule states that electrons will fill empty orbitals in the same sublevel before filling half-filled orbitals. This lowest-energy arrangement has the maximum number of unpaired electrons with parallel spins. The correct electron configuration for nitrogen is shown in (a), which is contrasted to an incorrect configuration shown in (b). The arrows in the $2p$ orbitals of configuration (a) could alternatively all point down.



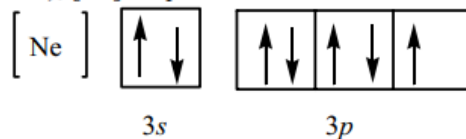
- 8.18 For elements in the same group (vertical column in periodic table), the electron configuration of the outer electrons are identical except for the n value. For elements in the same period (horizontal row in periodic table), their configurations vary because each succeeding element has one additional electron. The electron configurations are similar only in the fact that the same level (principal quantum number) is the outer level.
- 8.20 The total electron capacity for an energy level is $2n^2$, so the $n = 4$ energy level holds a maximum of $2(4^2) = 32$ electrons. A filled $n = 4$ energy level would have the following configuration: $4s^2 4p^6 4d^{10} 4f^{14}$.
- 8.21 Plan: Assume that the electron is in the ground state configuration and that electrons fill in a p_x - p_y - p_z order. By convention, we assign the first electron to fill an orbital an $m_s = +1/2$. Also by convention, $m_l = -1$ for the p_x orbital, $m_l = 0$ for the p_y orbital, and $m_l = +1$ for the p_z orbital.
Solution:
 a) The outermost electron in a rubidium atom would be in a $5s$ orbital. The quantum numbers for this electron are $n = 5$, $l = 0$, $m_l = 0$, and $m_s = +1/2$.
 b) The S^- ion would have the configuration $[\text{Ne}]3s^2 3p^5$. The electron added would go into the $3p_z$ orbital. Quantum numbers are $n = 3$, $l = 1$, $m_l = +1$, and $m_s = -1/2$.
 c) Ag atoms have the configuration $[\text{Kr}]5s^1 4d^{10}$. The electron lost would be from the $5s$ orbital with quantum numbers $n = 5$, $l = 0$, $m_l = 0$, and $m_s = +1/2$.
 d) The F atom has the configuration $[\text{He}]2s^2 2p^5$. The electron gained would go into the $2p_z$ orbital. Quantum numbers are $n = 2$, $l = 1$, $m_l = +1$, and $m_s = -1/2$.
- 8.23 a) Rb: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^1$
 b) Ge: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^2$
 c) Ar: $1s^2 2s^2 2p^6 3s^2 3p^6$
- 8.25 a) Cl: $1s^2 2s^2 2p^6 3s^2 3p^5$
 b) Si: $1s^2 2s^2 2p^6 3s^2 3p^2$
 c) Sr: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2$

8.27 Valence electrons are those electrons beyond the previous noble gas configuration and unfilled *d* and *f* sublevels.

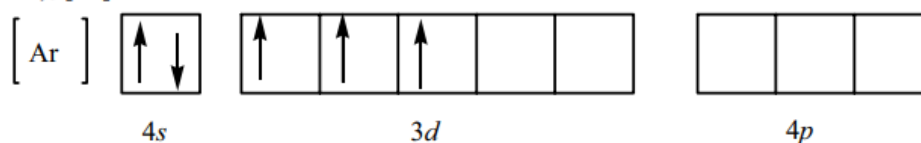
a) Ti ($Z = 22$); $[\text{Ar}]4s^23d^2$



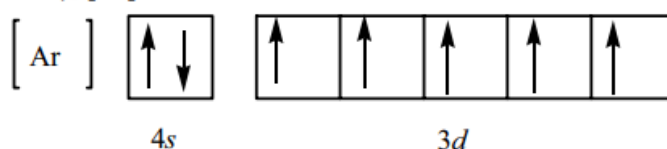
b) Cl ($Z = 17$); $[\text{Ne}]3s^23p^5$



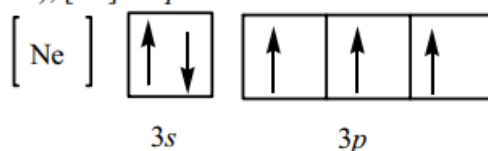
c) V ($Z = 23$); $[\text{Ar}]4s^23d^3$



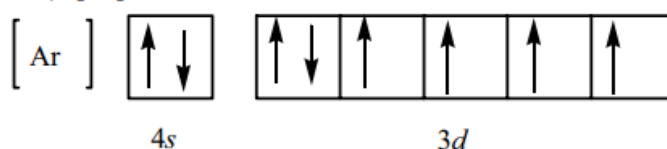
8.29 a) Mn ($Z = 25$); $[\text{Ar}]4s^23d^5$



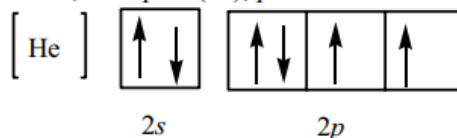
b) P ($Z = 15$); $[\text{Ne}]3s^23p^3$



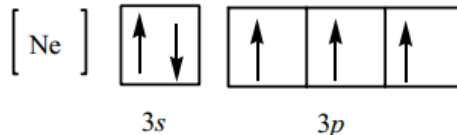
c) Fe ($Z = 26$); $[\text{Ar}]4s^23d^6$



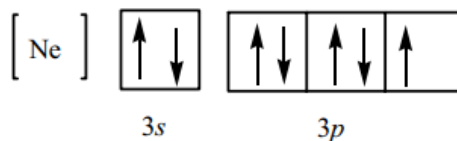
8.31 a) Element = O, Group 6A(16), period 2



b) Element = P, Group 5A(15), period 3

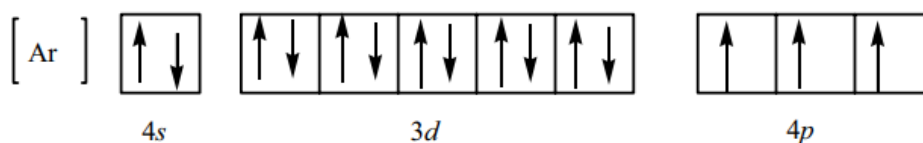


8.33 a)



Cl; Group 7A(17); period 3

b)



As; Group 5A(15); period 4

- 8.35 a) The orbital diagram shows the element is in period 4 ($n = 4$ as outer level). The configuration is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^1$ or $[\text{Ar}]4s^2 3d^{10} 4p^1$. One electron in the p level indicates the element is in group **3A(13)**. The element is Ga.
 b) The orbital diagram shows the $2s$ and $2p$ orbitals filled which would represent the last element in period 2, Ne. The configuration is $1s^2 2s^2 2p^6$ or $[\text{He}]2s^2 2p^6$. Filled s and p orbitals indicate the group **8A(18)**.
- 8.37 It is easiest to determine the types of electrons by writing a condensed electron configuration.
 a) O ($Z = 8$); $[\text{He}]2s^2 2p^4$. There are **2** inner electrons (represented by $[\text{He}]$) and **6** outer electrons. The number of valence electrons (**6**) equals the outer electrons in this case.
 b) Sn ($Z = 50$); $[\text{Kr}]5s^2 4d^{10} 5p^2$. There are 36 (from $[\text{Kr}]$) + 10 (from the filled d level) = **46** inner electrons. There are **4** outer electrons (highest energy level is $n = 5$) and **4** valence electrons.
 c) Ca ($Z = 20$); $[\text{Ar}]4s^2$. There are **2** outer electrons, **2** valence electrons, and **18** inner electrons.
 d) Fe ($Z = 26$); $[\text{Ar}]4s^2 3d^6$. There are **2** outer electrons (from $n = 4$ level), **8** valence electrons (the d orbital electrons count in this case because the sublevel is not full), and **18** inner electrons.
 e) Se ($Z = 34$); $[\text{Ar}]4s^2 3d^{10} 4p^4$. There are **6** outer electrons ($2 + 4$ in the $n = 4$ level), **6** valence electrons (filled d sublevels count as inner electrons), and **28** ($(18 + 10)$ or $(34 - 6)$) inner electrons.
- 8.39 a) The electron configuration $[\text{He}]2s^2 2p^1$ has a total of 5 electrons ($3 + 2$ from He configuration) which is element boron with symbol **B**. Boron is in group 3A(13). Other elements in this group are **Al, Ga, In, and Tl**.
 b) The electrons in this element total 16, 10 from the neon configuration plus 6 from the rest of the configuration. Element 16 is sulfur, **S**, in group 6A(16). Other elements in group 6A(16) are **O, Se, Te, and Po**.
 c) Electrons total $3 + 54$ (from xenon) = 57. Element 57 is lanthanum, **La**, in group 3B(3). Other elements in this group are **Sc, Y, and Ac**.
- 8.46 Atomic size increases down a main group and decreases across a period. Ionization energy decreases down a main group and increases across a period. These opposite trends result because as the atom gets larger, the outer electron is further from the attraction of the positive charge of the nucleus, which is what holds the electron in the atom. It thus takes less energy (lower IE) to remove the outer electron in a larger atom than to remove the outer electron in a smaller atom. As the atomic size decreases across a period due to higher Z_{eff} , it takes more energy (higher IE) to remove the outer electron.
- 8.48 For a given element, successive ionization energies always increase. As each successive electron is removed, the positive charge on the ion increases, which results in a stronger attraction between the leaving electron and the ion.
- When a large jump between successive ionization energies is observed, the subsequent electron must come from a lower energy level. Thus, by looking at a series of successive ionization energies, we can determine the number of valence electrons. For instance, the electron configuration for potassium is $[\text{Ar}]4s^1$. The first electron lost is the one from the $4s$ level. The second electron lost must come from the $3p$ level, and hence breaks into the core electrons. Thus, we see a significant jump in the amount of energy for the second ionization when compared to the first ionization.
- 8.53 a) Increasing atomic size: **K < Rb < Cs**, these three elements are all part of the same group, the alkali metals. Atomic size increases down a main group (larger outer electron orbital), so potassium is the smallest and cesium is the largest.
 b) Increasing atomic size: **O < C < Be**, these three elements are in the same period and atomic size decreases across a period (increasing effective nuclear charge), so beryllium is the largest and oxygen the smallest.
 c) Increasing atomic size: **Cl < S < K**, chlorine and sulfur are in the same period so chlorine is smaller since it is further to the right in the period. Potassium is the first element in the next period so it is larger than either Cl or S.
 d) Increasing atomic size: **Mg < Ca < K**, calcium is larger than magnesium because Ca is further down the alkaline earth metal group on the periodic table than Mg. Potassium is larger than calcium because K is further to the left than Ca in period 4 of the periodic table.
- 8.55 a) **Ba < Sr < Ca** The “group” rule applies in this case. Barium’s outer electron receives the most shielding; therefore, it is easiest to remove and has the lowest IE.
 b) **B < N < Ne** These elements have the same n , so the “period” rule applies. B experiences the lowest Z_{eff} and has the lowest IE. Ne has the highest IE, because it’s very difficult to remove an electron from the stable noble gas configuration.
 c) **Rb < Se < Br** IE decreases with increasing atomic size, so Rb (largest atom) has the smallest IE. Se has a lower IE than Br because IE increases across a period.
 d) **Sn < Sb < As** IE decreases down a group, so Sn and Sb will have smaller IE’s than As. The “period” rule applies for ranking Sn and Sb.

- 8.57 The successive ionization energies show a significant jump between the first and second IE's and between the third and fourth IE's. This indicates that 1) the first electron removed occupies a different orbital than the second electron removed, 2) the second and third electrons occupy the same orbital, 3) the third and fourth electrons occupy different orbitals, and 4) the fourth and fifth electrons occupy the same orbital. The electron configurations for period 2 elements range from $1s^2 2s^1$ for lithium to $1s^2 2s^2 2p^6$ for neon. The three different orbitals are $1s$, $2s$, and $2p$. The first electron is removed from the $2p$ orbital and the second from the $2s$ orbital in order for there to be a significant jump in the ionization energy between the two. The third electron is removed from the $2s$ orbital while the fourth is removed from the $1s$ orbital since IE_4 is much greater than IE_3 . The configuration is $1s^2 2s^2 2p^1$ which represents the five electrons in boron, **B**.
- 8.61 Three of the ways that metals and nonmetals differ are: 1) metals conduct electricity, nonmetals do not; 2) when they form stable ions, metal ions tend to have a positive charge, nonmetal ions tend to have a negative charge; and 3) metal oxides are ionic and act as bases, nonmetal oxides are covalent and act as acids. How many other differences are there?
- 8.62 Metallic character increases down a group and decreases toward the right across a period. These trends are the same as those for atomic size and opposite those for ionization energy.